14

Resonant Rydberg–Rydberg collisions

Resonant energy transfer collisions, those in which one atom or molecule transfers only internal energy, as oppposed to translational energy, to its collision partner require a precise match of the energy intervals in the two collision partners. Because of this energy specificity, resonant collisional energy transfer plays an important role in many laser applications, the He–Ne and CO_2 lasers being perhaps the best known examples.[1–4] It is interesting to imagine an experiment in which we can tune the energy of the excited state of atom B through the energy of the excited state of atom A, as shown in Fig. 14.1.[5] At resonance we would expect the cross section for collisionally transferring the energy from an excited A atom to a ground state B atom to increase sharply as shown in Fig. 14.1. In general, atomic and molecular energy levels are fixed, and the situation of Fig. 14.1 is impossible to realize. Nonetheless systematic studies of resonant energy transfer have been carried out by altering the collision partner, showing the importance of resonance in collisional energy transfer.[6–8]

The use of atomic Rydberg states, which have series of closely spaced levels, presents a natural opportunity for the study of resonant collisional energy transfer. One of the earliest experiments was the observation of resonant rotational to electronic energy transfer from NH_3 to Xe Rydberg atoms by Smith *et al.*[9] Resonant electronic to vibrational energy transfer has also been observed, from Rydberg states of Na to CH_4 and CD_4.[10] Both of these experiments, which are described in Chapter 11, correspond to tuning in steps equal to the discrete spacing of the Rydberg levels. In an ideal study of resonant collisions it would be

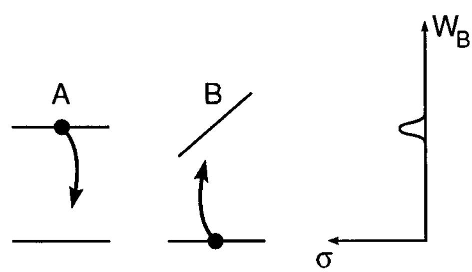

Fig. 14.1 Two atoms, A and B have energy levels as shown. Initially atom A is in its excited state, and atom B is in its ground state. If the energy of the excited state of atom B could be tuned by some means, we would expect the cross section for resonant energy transfer from atom A to atom B to increase at resonance as shown on the right (from ref. 5).

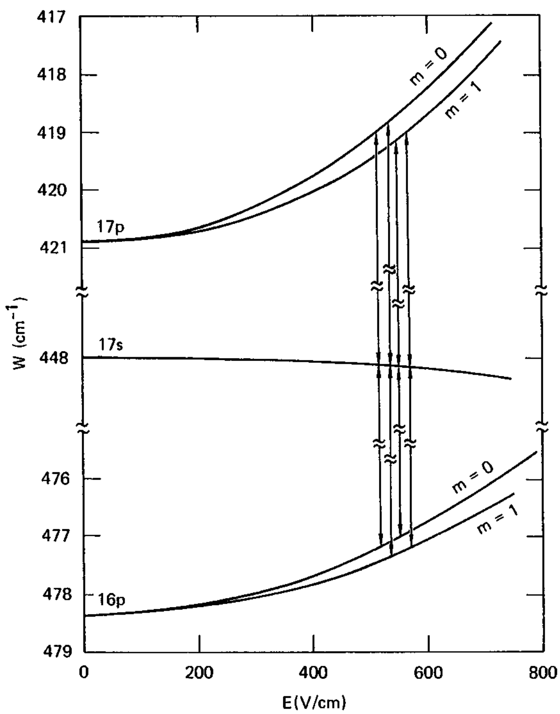

Fig. 14.2 Energy level diagram for the Na 16p, 17s, and 17p states in a static electric field. The vertical lines are drawn at the four fields where the s state is midway between the two p states and the resonant collisional transfer occurs (from ref. 12).

possible to tune continuously, as shown in Fig. 14.1, to observe the collisional resonance explicitly. In fact, due to the large Stark shifts of Rydberg atoms it is possible to tune them through a collisional resonance. A good example is the resonant energy transfer in the collision of two Na *n*s atoms in an electric field.[11] In Fig. 14.2 we show the energy levels of the Na *n*s, $(n-1)$p, and *n*p states as a function of electric field. In zero field the *n*s state lies slightly above midway between the two p states. However, as the field is increased the p states are shifted to higher energies by their Stark shifts. As shown by Fig. 14.2, the field also lifts the degeneracy of the $|m| = 0$ and 1 levels of the p states. Here m is the azimuthal orbital angular momentum quantum number. Due to the lifting of the m degeneracy there are four fields at which the *n*s state lies halfway between the two p states. At these fields two *n*s atoms can collide to produce an $(n-1)$p and an *n*p atom by the process

$$\mathrm{Na}\, ns + \mathrm{Na}\, ns \rightarrow \mathrm{Na}(n-1)\mathrm{p} + \mathrm{Na}\, n\mathrm{p}. \qquad (14.1)$$

As shown by Fig. 14.2, there are four collisional resonances for each *n*s state. We label them by (m_ℓ, m_u), where m_ℓ and m_u are the $|m|$ values of the lower and upper final p states. As shown by Fig. 14.2 in order of increasing field the

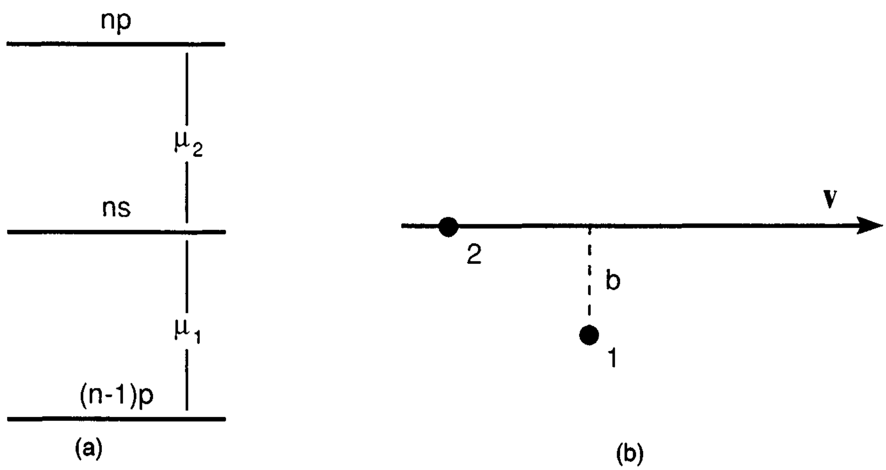

Fig. 14.3 (a) Energy levels for the Na ns, np and $(n-1)$p states showing the dipole matrix elements coupling them. (b) Diagram of the geometry of the collision in which atom 2 passes by atom 1 with velocity **v** and impact parameter b (from ref. 5).

resonances are (0,0), (1,0), (0,1), and (1,1). The Na collision process described in Eq. (14.1) and depicted in Fig. 14.2 is a resonant dipole–dipole process, and a simple picture enables us to determine both the magnitude of the peak cross section and the width of the collisional resonance. We consider the collision of two Na atoms initially in the ns state. During the collision they undergo transitions to the np and $(n-1)$p states, by means of dipole transitions with the dipole matrix elements μ_1 and μ_2, as shown by Fig. 14.3. We assume that the atoms follow undeflected straight line trajectories described by the impact parameter b and collision velocity **v**, as shown in Fig. 14.3. For simplicity, we first consider the special case of the static field's being tuned to the collisional resonance, i.e. to the field at which the energy of the ns state is midway between those of the np and $(n-1)$p states. We wish to calculate the probability of the atoms' making the transition to the np and $(n-1)$p states during the collision. One approach is to treat one atom, atom 2, as providing an oscillating field which drives the transition in the other atom, atom 1.[12] In other words, we treat atom 2 as a classical dipole. Classically, a dipole transition matrix element becomes a dipole of strength equal to the transition matrix element oscillating at the transition frequency. Thus atom 2 can be thought of as a collection of dipoles corresponding to the $n\text{s} \rightarrow n'\text{p}$ dipole transitions allowed from the ns state. Most of these transitions are off resonant when compared to the $n\text{s} \rightarrow (n-1)\text{p}$ transition, and we ignore them. The $n\text{s} \rightarrow (n-1)\text{p}$ transition in atom 2, while degenerate can be ignored on the basis of energy conservation (both atoms cannot make transitions to lower energy states). Thus the only important transition of atom 2 is the $n\text{s} \rightarrow n\text{p}$ transition, in which case the field produced by atom 2 at atom 1 is given by $E_2 \cong \mu_2/r^3$, μ_2 being the ns–np dipole matrix element, **r** the distance vector from atom 1 to atom 2, and

r the internuclear separation between the two atoms. This field is resonant with the $ns \rightarrow (n-1)$p transition of atom 1 and drives the transition if the interaction is strong enough. Since the dipole field falls quickly with increasing r, it is a reasonable approximation to set

$$E_2 = \mu_2/b^3 \qquad r \leqslant \frac{\sqrt{5}}{2}b \tag{14.2a}$$

$$E_2 = 0 \qquad r > \frac{\sqrt{5}}{2}b. \tag{14.2b}$$

This choice of cutoff implies that the field from atom 2 is present for a time $\tau = b/v$. In general atom 1 undergoes the $ns \rightarrow (n-1)$p transition if

$$E_2 \cdot \mu_1 \cdot \tau \approx 1, \tag{14.3}$$

or if

$$\frac{\mu_2\mu_1}{b^3} \cdot \frac{b}{v} = 1. \tag{14.4}$$

In Eqs. (14.3) and (14.4) μ_1 is the ns–$(n-1)$p dipole matrix element. Eq. (14.3) makes it apparent that the requirement for driving the transition is that the time integrated interaction be one. Solving Eq. (14.4) for b^2, ignoring factors of π, gives the cross section σ_R for the resonant collisional energy transfer. Explicitly,

$$\sigma_R \approx b^2 = \frac{\mu_1\mu_2}{v}. \tag{14.5}$$

Using the impact parameter b and the velocity v we can also calculate the duration, or time, of the collision,

$$\tau = \frac{b}{v} = \frac{\sqrt{\mu_1\mu_2}}{v^{3/2}}. \tag{14.6}$$

For the $ns \rightarrow n$p and $ns \rightarrow (n-1)$p transitions the dipole matrix elements are $\sim n^2$. Accordingly, we insert n^2 for μ_1 and μ_2 in Eqs. (14.5) and (14.6), finding

$$\sigma_R = n^4/v \tag{14.7}$$

and

$$\tau = n^2/v^{3/2}. \tag{14.8}$$

As shown by Eq. (14.7), the cross section is given by the geometric cross section divided by the collision velocity. For thermal collisions $v \sim 10^{-4}$, and as a result, $\sigma_R \sim 10^4 n^4$. For n=20 this expression yields a cross section of $\sim 10^9 a_0{}^2$, or $\sim 2 \times 10^8$ Å^2, which is large by atomic standards. Perhaps more interesting than the magnitudes of the cross sections are the collision times. Evaluating Eq. (14.8) for $n = 20$ and $v = 10^{-4}$ gives $\tau = 4 \times 10^8$ or $\sim 10^{-9}$ s, a time far longer than the typical atomic collision times of 10^{-12} s.

The discussion above is focused on dipole–dipole collision processes, but it may easily be extended to higher multipole processes. If atoms 1 and 2 have $2k$ pole moments of n^{2k} and $n^{2k'}$ respectively, when the atoms are separated by a distance r the interaction V due to these multipoles is

$$V = \frac{n^{2k} n^{2k'}}{r^{1+k+k'}}. \tag{14.9}$$

If we again assume that the atoms collide with an impact parameter b and velocity v, following straight line trajectories , we can again assume that V is only non zero for $r \approx b$ and a time interval $\tau = b/v$. Requiring that $V\tau = 1$ for a significant transition probability leads to

$$\sigma \cong b^2 = \frac{n^4}{v^{2/(k+k')}}. \tag{14.10}$$

For a dipole–dipole collision, $k = k' = 1$, and this result reduces to Eq. (14.5). On the other hand, as k and k' increase, the factor $v^{2/(k+k')}$ approaches 1 from below, and the cross section decreases to the geometric size of the atom. It is worth noting that the assumption of a straight line trajectory is not strictly valid in mixed multipole collisions, or in any collision in which the internal angular momentum of the atoms is not conserved. However, the amount of angular momentum in the translational motion is usually sufficiently high that small changes in it do not significantly alter the trajectories of the colliding atoms.

Two state theory

The expression of Eq. (14.7) for the cross section is adequate for a calculation of the magnitude of the cross section, but it does not take into account effects due to the orientation of the collision velocity $\mathbf{v}$ relative to the tuning field $\mathbf{E}$, nor does it allow the calculation of the lineshape of the collisional resonance. With these thoughts in mind, we present a more refined treatment of the problem which enables us to describe some of the details of the process. We shall consider the resonant collisions of two Na ns atoms to yield an np and an $(n-1)$p atom, as indicated by Eq. (14.1). The description is, in principle, similar to the treatment of resonant rotational energy transfer between polar molecules given by Anderson.[8] It differs, however, in that the presence of the electric field, which lowers the symmetry, must be taken into account. Treatments specific to the problem of dipole–dipole collisions in a field, such as the process depicted by Fig. 14.2, have been given by Gallagher *et al.*,[12] Fiordilino *et al.*,[13] and Thomson *et al.*[14]

We assume again that the atoms follow straight line trajectories, and we calculate the transition probability, $P(b)$, from the initial to the final state in a collision with a given impact parameter, b. We then compute the cross section by integrating over impact parameter, and, if necessary, angle of $\mathbf{v}$ relative to $\mathbf{E}$ to obtain the cross section. The central problem is the calculation of the transition probability $P(b)$. The Schroedinger equation for this problem has the Hamiltonian

$$H = H_0 + V, \tag{14.11}$$

where $H_0 = H_1 + H_2$ is the Hamiltonian for two non–interacting atoms in the static field, and V is the dipole–dipole interaction

$$V = \frac{\boldsymbol{\mu}_1 \cdot \boldsymbol{\mu}_2}{r^3} - \frac{3(\boldsymbol{\mu}_1 \cdot \mathbf{r})\,(\boldsymbol{\mu}_2 \cdot \mathbf{r})}{r^5}. \tag{14.12}$$

We construct the molecular product states which have the spatial wavefunctions

$$\begin{aligned} \psi_{\mathrm{A}} &= \psi_{1ns} \otimes \psi_{2ns} \\ \psi_{\mathrm{B}} &= \psi_{1np} \otimes \psi_{2(n-1)\mathrm{p}}, \end{aligned} \tag{14.13}$$

where $\psi_{1n\ell}$ and $\psi_{2n\ell}$ are the atomic wavefunctions of atoms 1 and 2 in the $n\ell$ state. These product states are the time independent solutions to H_0, having energies $W_{\mathrm{A}} = W_{ns} + W_{ns}$ and $W_{\mathrm{B}} = W_{np} + W_{(n-1)\mathrm{p}}$.

The dipole–dipole interaction V has a diagonal term, which slightly shifts the energies, and an off diagonal term, which induces the collisional transitions. Since V depends on r, V is time dependent, and the wavefunction is given by the solution to the time dependent Schroedinger equation

$$H\psi = \mathrm{i}\partial\psi/\partial t. \tag{14.14}$$

When the two possible states are those given by Eq. (14.13) the general form of the solution to Eq. (14.14) is

$$\psi = C_{\mathrm{A}}(t)\psi_{\mathrm{A}} + C_{\mathrm{B}}(t)\psi_{\mathrm{B}}, \tag{14.15}$$

in which all of the time dependence resides in the coefficients $C_{\mathrm{A}}(t)$ and $C_{\mathrm{B}}(t)$. Using the wavefunction of Eq. (14.15) in the time dependent Schroedinger equation, Eq. (14.14), we find two coupled equations for $C_{\mathrm{A}}(t)$ and $C_{\mathrm{B}}(t)$,

$$W_{\mathrm{A}}C_{\mathrm{A}}(t) + V_{\mathrm{AB}}C_{\mathrm{B}}(t) + V_{\mathrm{AA}}C_{\mathrm{A}}(t) = \mathrm{i}\dot{C}_{\mathrm{A}}(t), \tag{14.16a}$$

$$W_{\mathrm{B}}C_{\mathrm{B}}(t) + V_{\mathrm{BA}}C_{\mathrm{A}}(t) + V_{\mathrm{BB}}C_{\mathrm{B}}(t) = \mathrm{i}\dot{C}_{\mathrm{B}}(t), \tag{14.16b}$$

where $V_{\mathrm{BA}} = \langle\psi_{\mathrm{B}}|V|\psi_{\mathrm{A}}\rangle = \int\psi_{\mathrm{B}}^* V\psi_{\mathrm{A}}\,\mathrm{d}\tau_1\,\mathrm{d}\tau_2$ is the matrix element of V between the two spatial wavefunctions.

In Eqs. (14.16) the matrix elements of V are time dependent, since the internuclear distance is a function of time, as are the coefficients $C_{\mathrm{A}}(t)$ and $C_{\mathrm{B}}(t)$. In general, Eqs. (14.16) cannot be solved analytically, but in several special cases analytic solutions can be obtained.

Box interaction strength approximation

If we set $V_{\mathrm{AA}} = V_{\mathrm{BB}} = 0$ and use Eq. (14.2), i.e. assume that V_{AB} is a constant, $= \mu_1\mu_2/b^3$, for $r < \sqrt{5}b/2$ and zero for $r > \sqrt{5}b/2$, as given in Eq (14.2), the resulting approximation is equivalent to the usual molecular beam magnetic resonance treatment.[15] If we assume that $r = \sqrt{b^2+v^2t^2}$ and that initially both atoms are in the ns state so that $\psi(t_0) = \psi_A$, $C_A(t_0) = 1$ and $C_B(t_0) = 0$ for $t_0 < b/2v$. After the collision, $t > b/2v$, the probability of finding the atoms in the p states, $\psi = \psi_B$, is given by

$$P(b) = |C_B(t)|^2 = \frac{|V_{BA}|^2}{\Omega^2}\sin^2\left(\frac{\Omega b}{v}\right), \tag{14.17}$$

where $\Omega = \sqrt{(W_A - W_B)^2 + 4|V_{BA}|^2}/2$. Eq. (14.17) yields Lorentzian resonances with linewidths $\sim(\sqrt{v^3/\mu_1\mu_2})$, in agreement with Eq. (14.6). While the widths are meaningful, the lineshape, which depends on the form of the interaction during the collision, is only approximate.

At resonance $W_A = W_B$, and Eq. (14.17) reduces to

$$P(b) = \sin^2\left(\frac{\mu_1\mu_2}{b^2 v}\right), \tag{14.18}$$

in which we have used the explicit form $V_{AB} = \mu_1\mu_2/b^3$.

Exact resonance approximation

Another case which may be treated analytically is the case of exact resonance, $W_A = W_B$, if in addition, $V_{AA} = V_{BB} = 0$ and $V_{AB} = V_{BA}$, i.e. the coupling matrix element is real. In this case Eqs. (14.16) are readily decoupled, leading to two identical uncoupled equations. The equation for $C_A(t)$ is

$$\ddot{C}_A(t) = \frac{\dot{V}_{AB}}{V_{AB}}\dot{C}_A(t) - V_{AB}^2 C_A(t). \tag{14.19}$$

The equation for $C_B(t)$ has identical coefficients. If we again assume that initially, at $t = -\infty$, $C_A(-\infty) = 1$ and $C_B(-\infty) = 0$ the solutions to Eq. (14.19) which satisfy these boundary conditions are[12,13]

$$C_A(t) = \cos\left(\int_{-\infty}^{t} V_{AB}(t')\mathrm{d}t'\right) \tag{14.20a}$$

and

$$C_B(t) = \sin\left(\int_{-\infty}^{t} V_{AB}(t')\mathrm{d}t'\right). \tag{14.20b}$$

The probability of making a transition is therefore given by

$$P(\infty) = C_B^2(\infty) = \sin^2\left(\int_{-\infty}^{\infty} V_{AB}(t')\mathrm{d}t'\right)\cdot \tag{14.21}$$

To obtain an explicit form for the matrix element V_{AB} given in Eq. (14.12), requires that we define the geometry of the collision. While the collision velocity is normally the logical choice of quantization axis for field free collisions, the

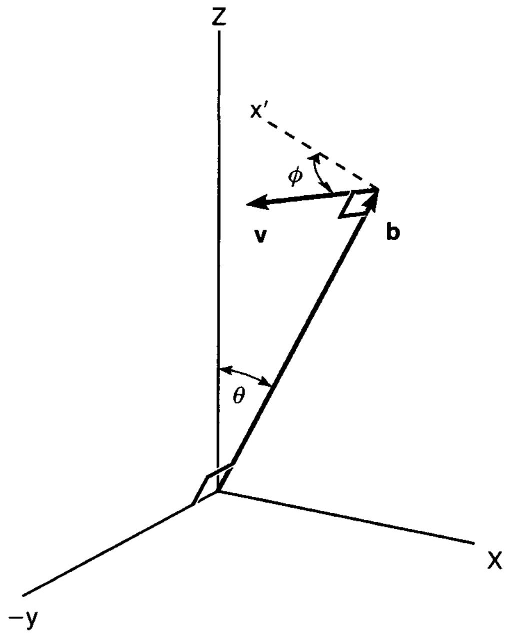

Fig. 14.4 Diagram of the geometry of the collision. The static field is in the z direction. The impact parameter vector **b** lies in the x,z plane, at an angle θ from the z axis. The velocity vector **v** lies in the plane perpendicular to **b** and makes an angle ϕ with the x' axis, which is perpendicular to **b** and in the x,z plane (from ref. 5).

presence of the tuning electric field makes the **E** field direction the logical choice, since the eigenstates of H_0 are easily described with this choice of quantization axis. If we define the field direction as the z axis, so that $\mathbf{E}\|\hat{\mathbf{z}}$, then the impact parameter vector **b** makes an angle θ with the z axis as shown in Fig. 14.4. We may assume, with no loss of generality, that **b** lies in the x,z plane. The collision velocity **v**, which is a constant vector due to the assumption of straight line trajectories, is normal to **b** and makes an angle ϕ with the x′ axis, which is in the x,z plane and perpendicular to **b**, as shown in Fig. 14.4. If the closest approach of the two atoms occurs at $t = 0$, the coordinates of atom 2 relative to atom 1 are given by

$$\begin{aligned} y &= -vt\sin\phi \\ x &= b\sin\theta - vt\cos\phi\cos\theta \\ z &= b\cos\theta + vt\cos\phi\sin\theta. \end{aligned} \tag{14.22}$$

Using these coordinate values we may now evaluate the matrix elements of Eq. (14.12) by substituting for the dipole moments the dipole matrix elements between the initial and final states. This procedure yields explicitly time dependent matrix elements $V_{AB}(t)$. It is particularly interesting to consider the (0,0) resonances, for two reasons. First, the (0,0) resonances have no further splitting due to the spin orbit interaction and are therefore good candidates for detailed experimental study. Second, since these resonances only involve the matrix elements of μ_z, which are real, it is possible to evaluate the transition probability at resonance analytically using Eq. (14.21). The matrix element $V_{AB}(t)$ is given by

$$V_{AB}(t) = \mu_{z_1}\mu_{z_2}\left(\frac{1}{r^3} - \frac{3b^2\cos^2\theta}{r^5} - \frac{6bvt\cos\theta\cos\phi}{r^5} - \frac{3v^2t^2\cos^2\phi\sin^2\theta}{r^5}\right), \quad (14.23)$$

where $r = \sqrt{b^2 + v^2t^2}$.

In Fig. 14.5 we show graphs of $V_{AB}(t)$ for two cases; $\mathbf{v}\|\mathbf{E}$, and one example of $\mathbf{v} \perp \mathbf{E}$, for which $\theta = 0$ and $\phi = \pi/2$. Note that the specification $\mathbf{v} \parallel \mathbf{E}$, specifies both θ and ϕ. However $\mathbf{v} \perp \mathbf{E}$, only specifies $\phi = \pi/2$; θ can take any value.

The form of $V_{AB}(t)$ given in Eq. (14.23) can be integrated analytically to find the transition probability at resonance. Carrying out the integration yields

$$\int_{-\infty}^{\infty} V_{AB}(t')\mathrm{d}t' = \frac{2\mu_{z_1}\mu_{z_2}}{vb^2}(1 - 2\cos^2\theta - \cos^2\phi\sin^2\theta). \quad (14.24)$$

Evaluating Eq. (14.24) for $\mathbf{v} \parallel \mathbf{E}$, $\theta = \pi/2$ and $\phi = 0$, reveals that the integral vanishes, as might have been anticipated from a close examination of Fig. 14.5(a). The average value of V_{AB} in Fig. 14.5(a) is clearly near zero. On the other hand, for $\mathbf{v} \perp \mathbf{E}$, $\theta = 0$ and $\phi = \pi/2$, the integral of Eq. (14.24) does not vanish, since the average value of $V_{AB}(t)$ does not vanish in this case, as can be seen in Fig. 14.5(b). Correspondingly, at resonance, the cross section must vanish for $\mathbf{v} \parallel \mathbf{E}$ but not for $\mathbf{v} \perp \mathbf{E}$.

Numerical calculations

Normally one might expect that if the transition probability vanishes on resonance it also vanishes off resonance. However, such is not the case. When the transition probability is calculated off resonance, by numerically solving Eqs. (14.16) using a Taylor expansion method, it is nonzero for both $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$.[14,16] In Fig. 14.6 we show the transition probabilities obtained using two different approximations for $\mathbf{v} \parallel \mathbf{E}$, and $\mathbf{v} \perp \mathbf{E}$ for the 17s (0,0) collisional resonance.[16] To allow direct comparison to the analytic form of Eq. (14.21) we show the transition probabilities calculated with $V_{AA} = V_{BB} = 0$. For these calculations the parameters $\mu_{z_1} = \mu_{z_2} = 156.4\ ea_0$, $b = 10^4 a_0$, and $v = 1.6 \times 10^{-4}$ au have been used. The resulting transition probability curves are shown by the broken lines of Fig. 14.6. As shown by Fig. 14.6 these curves are symmetric about the resonance position. The $\mathbf{v} \perp \mathbf{E}$ curve of Fig. 14.6(b) has an approximately Lorentzian form, but the $\mathbf{v} \parallel \mathbf{E}$ curve of Fig. 14.6(a), while it vanishes on resonance as predicted by Eq. (14.24), has an unusual double peaked structure.

Since the transition probabilities of Fig. 14.6 are obtained numerically, it is not appreciably more difficult to include the diagonal matrix elements V_{AA} and V_{BB} arising from the permanent dipole moments of the atomic states in the field. The

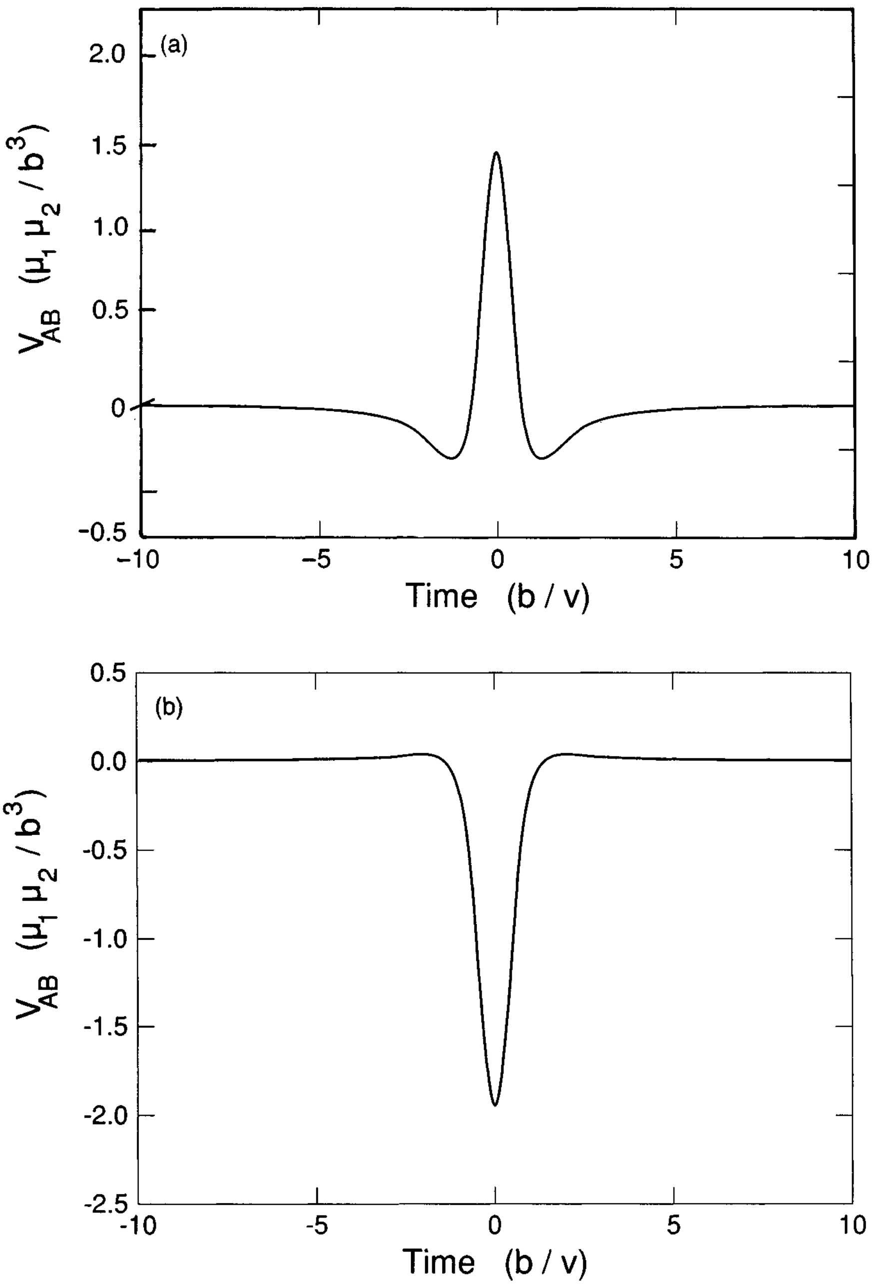

Fig. 14.5 Dipole–dipole interaction potential for (a) the case in which $\mathbf{v} \parallel \mathbf{E}$, (b) the case in which $\mathbf{v} \perp \mathbf{E}$ with $\theta = 0$ and $\phi = \pi/2$ (from ref. 14).

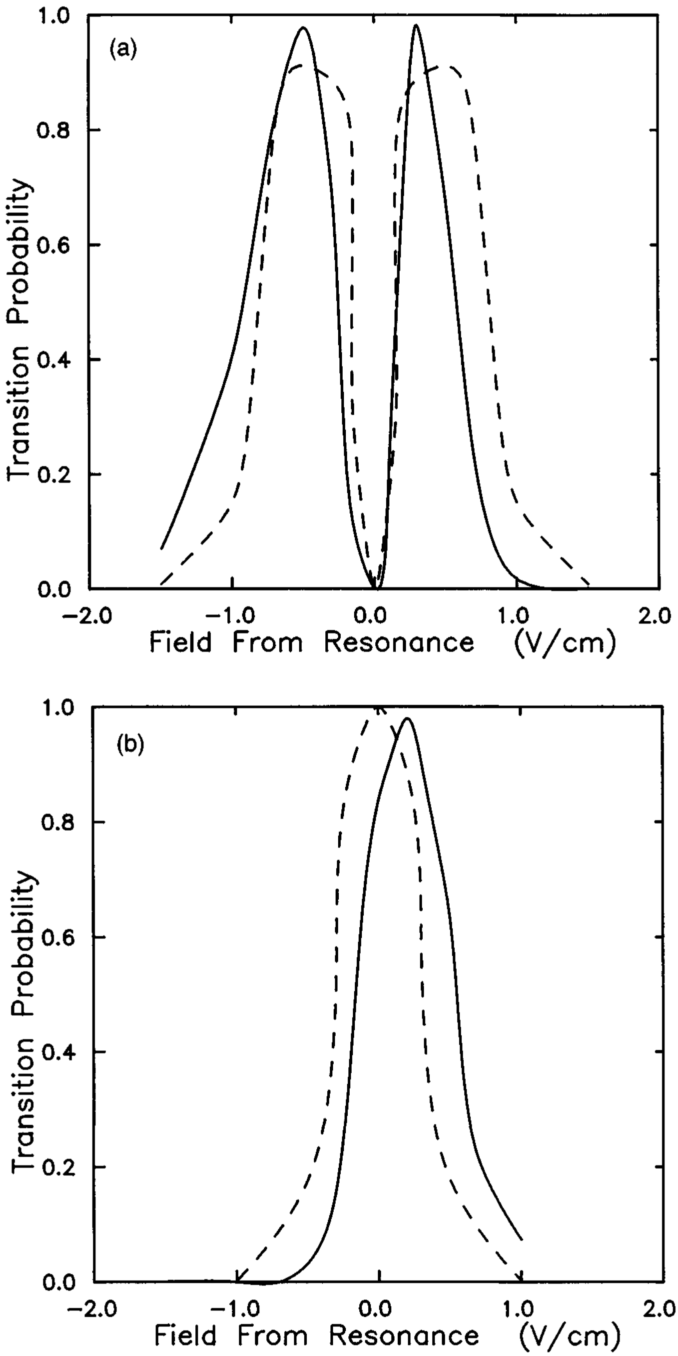

Fig. 14.6 Calculations of the transition probability for the Na 17s + 17s → 17p + 16p, (0,0) transition for (a) $\mathbf{v} \parallel \mathbf{E}$, and (b) $\mathbf{v} \perp \mathbf{E}$. Solid lines are for calculations which include the effects of the permanent dipole moments; dashed lines indicate that the permanent dipole moments have been neglected (from ref. 16).

solid lines of Fig. 14.6 are the transition probabilities calculated with the permanent dipole moments at the field of the collisional resonance taken into account. The permanent dipole moments of the 16p, 17p, and 17s states are $\mu_{16p} = 113.3ea_0$, $\mu_{17p} = 182.3\ ea_0$, and $\mu_{17s} = -16.5\ ea_0$, respectively. As shown by Fig. 14.6, when the permanent moments are taken into account the location of the resonance shifts slightly, by approximately a resonance linewidth, as expected from the fact that the permanent moments are nearly as large as the transition moments. In addition, the lineshapes become slightly asymmetric, due to the fact that the energy spacing of the atomic states now changes with internuclear separation.

Intracollisional interference

The unusual lineshape of Fig. 14.6(a) for $\mathbf{v} \parallel \mathbf{E}$ has approximately the same form and origin as the Ramsey separated oscillatory field pattern when there is a 180° phase shift between the two oscillatory fields.[14] To see the similarity we first examine $V_{AB}(t)$, shown in Fig 14.5(a). As shown by Fig. 14.5(a), for $\mathbf{v} \parallel \mathbf{E}$ V_{AB} changes sign twice during the collision, so that $\int_{-\infty}^{\infty} V_{AB}(t')\mathrm{d}t' = 0$. A sign change of $V_{AB}(t')$ is equivalent to a 180° phase shift between the oscillatory fields in a separated oscillatory field experiment, and the $\mathbf{v} \parallel \mathbf{E}$ collisions are thus approximately equivalent to doing an rf resonance experiment in which there are three sequential rf fields with phase changes of 180° between successive fields. At resonance, the contributions to the transition amplitude from $V_{AB}(t') < 0$ cancel those from $V_{AB}(\mathrm{t}') > 0$ so that, integrated over the whole collision, the transition probability vanishes. Off resonance, however, the cancellation is not complete.

Calculation of cross sections

To convert the transition probabilities to cross sections we must integrate over impact parameter using

$$\sigma = \int_0^\infty P(b) 2\pi b \mathrm{d}b. \tag{14.25}$$

It is useful to consider the case of exact resonance, which may be worked out analytically. At resonance, using either Eq. (14.18) or Eq. (14.21) the transition probability is given by

$$P(b) = \sin^2\left(\frac{D}{b^2 v}\right), \tag{14.26}$$

where $D = \mu_1\mu_2$ if we use Eq. (14.18) and $D = 2\mu_{z_1}\mu_{z_2}$ $(1 - 2\cos^2\theta - \cos^2\phi\sin^2\theta)$ if we use Eq. (14.21). In either case D is proportional to the product of the two dipole moments. Examining Eq. (14.26) we see that if we define b_0 such that $D/b_0{}^2v = \pi/2$, as b increases from 0 to b_0, $P(b)$ oscillates between 0 and 1, and as b increases from b_0 to infinity $P(b)$ decreases smoothly to zero. If we insert the transition probability of Eq. (14.26) into Eq. (14.25), the integral can be done analytically, yielding

$$\sigma = \frac{\pi^3}{4}b_0{}^2 = \frac{\pi^2}{2}\frac{D}{v}. \tag{14.27}$$

Recalling that D $\sim \mu_1\mu_2$, we see that Eq. (14.27) resembles Eq. (14.5) which was derived in a very simple way.

If we are interested in the cross section for $\mathbf{v} \parallel \mathbf{E}$, the angles θ and ϕ are uniquely defined, $\theta = \pi/2$ and $\phi = 0$. On the other hand, to calculate the observed $\mathbf{v} \perp \mathbf{E}$ cross section Eq. (14.27) must be averaged over θ for $\phi = \pi/2$. In either case the calculated cross sections must be averaged over the appropriate distribution of collision velocities.

In an analogous fashion the numerically obtained transition probabilities shown in Fig. 14.6 can be converted to cross sections by integrating over impact parameter and averaging over the possible collision velocities. Collisions with $\mathbf{v} \perp \mathbf{E}$ do not have a single allowed value of θ. However, since the lineshape of Fig. 14.6(b) is simple, some averaging over the possible values of θ has no appreciable effect. For $\mathbf{v} \parallel \mathbf{E}$, θ and ϕ are fixed, at 90° and 0°, so the integration over impact parameter is only over b, and the calculated $\mathbf{v} \parallel \mathbf{E}$ cross section is very similar to Fig. 14.6.

The lineshapes of Figs. 14.6(a) and 14.6(b) are very different. However, this difference is only apparent in an experiment with high resolution. At low resolution the $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ collisional resonances can be expected to exhibit the same instrumentally determined shape. Furthermore, both cross sections are approximately the same size. The calculated peak cross sections for the Na 17s + 17s $\rightarrow$ 16p + 17p collisional resonances are $6.0 \times 10^8 a_0{}^2$ and $1.0 \times 10^9 a_0{}^2$ for $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ respectively.[14] When integrated over the tuning field, to allow comparison to low resolution experiments, the two integrated cross sections are $8.3 \times 10^8 a_0{}^2$ V/cm and $1.2 \times 10^9 a_0{}^2$ V/cm.[14]

Experimental approach

The basic principle of the experimental approach is easily understood by considering the Na collisions of Eq. (14.1) and Fig. 14.2 as a concrete example. As shown by Fig. 14.7, atoms effuse from a heated oven in which the Na vapor pressure is ~1 Torr.[14] The Na atoms are collimated into a beam by a collimator, not shown in

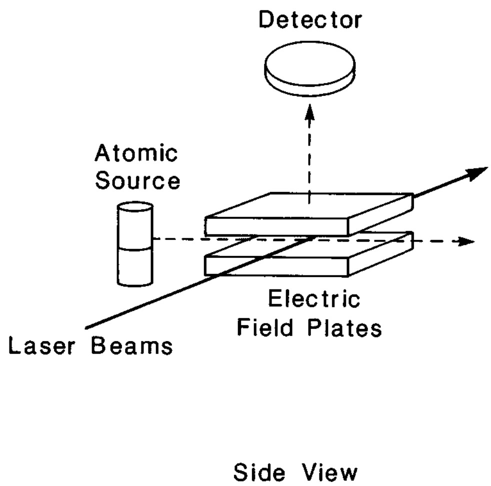

Fig. 14.7 Experimental apparatus configured with the electric field perpendicular to the collisional velocity (from ref. 14).

Fig. 14.7, and pass into the interaction region, between a pair of electric field plates, at which point the beam density can be up to 10^9 cm^{-3}. The Na atoms are excited to an ns state by two 5 ns pulsed dye lasers tuned to the Na 3s$\rightarrow$3p and 3p$\rightarrow$ *n*s transitions at 5890 Å and ~4150 Å respectively. The *n*s atoms are allowed to collide with each other for a period of ~1 μs during which time the slow atoms in the beam are overtaken by the fast atoms. Thus the collision velocity is approximately the width of the velocity distribution of the beam. After the 1 μs period during which the atoms collide a high voltage pulse is applied to the lower plate of the interaction region. The amplitude of the pulse is set to ionize atoms in the *n*p state, the higher lying final state of the collision, but not atoms in the *n*s state, the initial state of the collision. Ions produced by field ionization are expelled from the region between the plates and impinge upon a particle multiplier above the interaction region. The signal from the particle multiplier is recorded with a gated integrator. A static tuning voltage is applied to the lower plate to tune the atomic levels into resonance. The collisional resonances are observed by monitoring the field ionization signal from atoms in the *n*p state as the tuning voltage is slowly swept through the collisional resonances. A typical example is shown in Fig. 14.8, a recording of the field ionization signal from the Na 17p state when the 17s state is populated by laser excitation.[12]

n scaling laws

One of the most striking aspects of resonant collisional energy transfer between Rydberg atoms is the magnitude of the cross sections. Accordingly, the first

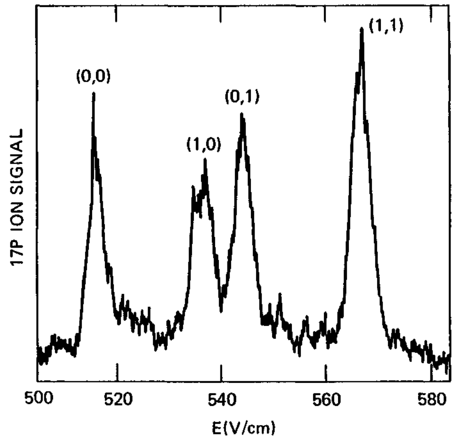

Fig. 14.8 The observed Na 17p ion signal after population of the 17s state vs dc electric field, showing the sharp collisional resonances. The resonances are labeled by the |m| values of the lower and upper p states (from ref. 12).

experiments were focused on determining the magnitude of the cross sections and verifying the n scaling laws for the Na ns + ns → np + (n − 1)p cross sections and collision times.

The cross sections were measured by taking advantage of the fact that the number of atoms, N_p, which are in the np state after the atoms have been allowed to collide for a time interval T is given by[12]

$$N_p = \frac{N_s^2 \sigma \bar{v} T}{V}, \tag{14.28}$$

where N_s is the number of atoms in the ns state, V is the volume of the sample of excited atoms, $\bar{v}$ is the average collision velocity, and σ is the cross section. Eq. (14.28) is valid only if the population of the ns state is not depleted significantly. If significant depletion of the population in the ns state occurs, corrections to Eq. (14.28) must be introduced. The number of atoms in the np state, N_p, is related to the signal associated with np state, N_p', by the conversion efficiency of the detector, Γ. Specifically $N_p' = \Gamma N_p$, and $N_s' = \Gamma N_s$. Using N_p' and N_s' we can rewrite Eq. (14.28) as

$$\sigma = \frac{N_p'}{(N_s')^2}\left[\frac{\bar{v}T\Gamma}{V}\right]. \tag{14.29}$$

If the geometry of the experiment, timing gates, oven temperature, and detector gain are fixed, then the quantity in the square brackets of Eq. (14.29) is constant, and the relative cross section as a function of n is easily obtained by measuring the signals N_s' and N_p' as a function of n. The relative cross sections were put on an absolute basis by measuring the quantities in the brackets of Eq. (14.29), with the result shown in Fig. 14.9. The magnitudes and the n scaling of the

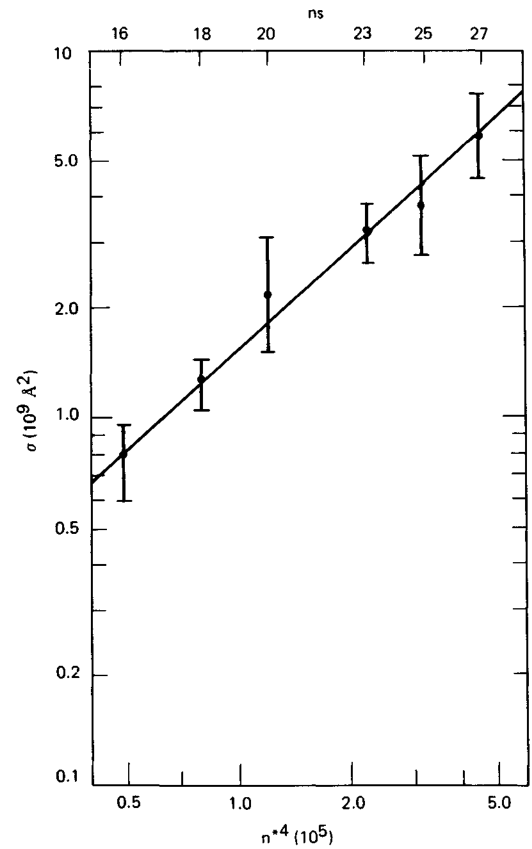

Fig. 14.9 The observed cross sections with their relative error bars (●), and the fit curve (—) (from ref. 12).

cross section as $n^{*3.7(5)}$ are in good agreement with Eq. (14.7).[12] Here n^* is the effective quantum number of the ns state, defined by $W_{ns} = -1/2n^{*2}$. The collisions resulting in the cross sections shown in Fig. 14.9 were most likely not between atoms of different velocities in a directed beam but between atoms moving in random directions. Both the excited atom sample volume and collision velocity were probably a factor of 2 larger than thought at the time, so the errors compensated to a large extent. In any event, from Fig. 14.9 it is apparent that the cross section exhibits the expected n^4 scaling and is of the correct magnitude to match the expected $\sigma = n^4/v$ behavior.

Verification that the collision time τ increases as n^2, or that the linewidth of the collisional resonances decreases as n^{-2}, is simply a matter of measuring the widths of the collisional resonances. In Fig. 14.10 we show a plot of the width of the (0,0) resonance as a function of n. The observed widths exhibit magnitudes and an $n^{*-1.95(20)}$ dependence in agreement with Eq. (14.8).

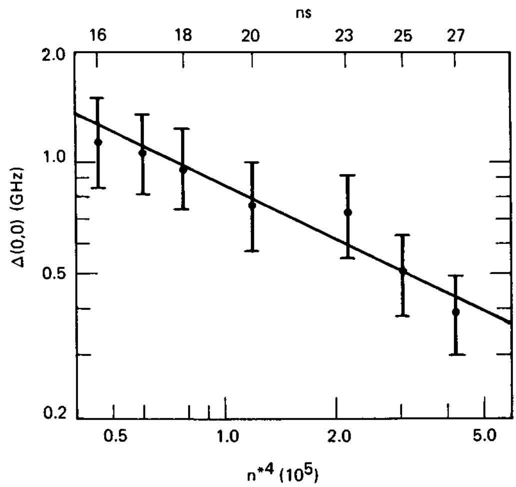

Fig. 14.10 The widths of the (0,0) resonances (●), and the fit curve (—) (from ref. 12).

Orientation dependence of v and E and intracollisional interference

One of the more interesting aspects of the collision process is the calculated dependence on the orientation of **v** relative to **E**. In particular, do the lineshapes for collisions with $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ differ as dramatically as shown in Fig. 14.6? To answer this question unambiguously required two improvements upon the initial measurements. First, the Na atoms must be in a well defined beam, otherwise **v** is not well defined. This requirement is easily met by enclosing the interaction region with a liquid N_2 cooled box, which ensures that the only Na atoms in the interaction region are those in the beam. Second, the field homogeneity must be adequate, ~1 part in 10^4, to resolve the intrinsic lineshape of the collisional resonances. A pair of Cu field plates 1.592(2) cm apart with 1 mm diameter holes in the top plate to allow the ions to be extracted is adequate to meet this requirement.

Making these two modifications, but keeping the basic geometry of Fig. 14.7, substantially improves the resolution so that what appear to be four collisional resonances in Fig. 14.8 are actually nine.[14] All the resonances but the (0,0) resonance are split by the spin orbit splitting of the np $|m| = 1$ states. In the (0,1) and (1,0) resonances the upper and lower p states, respectively, are split, leading to doublets, while in the (1,1) resonance both upper and lower p states are split, leading to a quartet. In view of the fact that only the (0,0) resonance is a single line, it was selected for a detailed investigation of the cross section with $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$. To measure the cross section with $\mathbf{v} \perp \mathbf{E}$ the geometry of Fig. 14.8, with the liquid N_2 cooled shield, is adequate. To measure the cross section with $\mathbf{v} \parallel \mathbf{E}$ an arrangement, shown in Fig. 14.11, is used in which the atomic beam passes

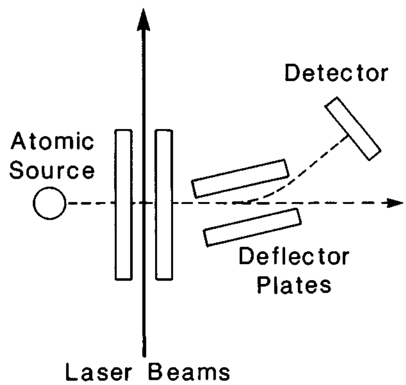

Fig. 14.11 Experimental apparatus configured so that the electric field and collisional velocity are parallel (from ref. 14).

through the field plates. In both cases the collision velocity is the average of the difference in velocities of different atoms in the beam.

The observed cross sections for the 18s (0,0) collisional resonance with $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ are shown in Fig. 14.12. The approximately Lorentzian shape for $\mathbf{v} \perp \mathbf{E}$ and the double peaked shape for $\mathbf{v} \parallel \mathbf{E}$ are quite evident. Given the existence of two experimental effects, field inhomogeneties and collision velocities not parallel to the field, both of which obscure the predicted zero in the $\mathbf{v} \parallel \mathbf{E}$ cross section, the observation of a clear dip in the center of the observed $\mathbf{v} \parallel \mathbf{E}$ cross section supports the theoretical description of intracollisional interference given earlier. It is also interesting to note that the observed $\mathbf{v} \parallel \mathbf{E}$ cross section of Fig. 14.12(a) is clearly asymmetric, in agreement with the transition probability calculated with the permanent electric dipole moments taken into account, as shown by Fig. 14.6.

Velocity dependence of the collisional resonances

One of the potentially most interesting aspects of the resonant collisions is that, in theory, the collision time increases and the linewidth narrows as the collision velocity is decreased. According to Eqs. (14.6) and (14.8) the collision time is proportional to $1/v^{3/2}$. Collisions between thermal atoms with temperatures of ~500 K lead to linewidths of the collisional resonances that are a few hundred MHz at $n = 20$. In principle, substantially smaller linewidths can be observed if the collision velocity is reduced.

The most straightforward way of reducing the collision velocity is to velocity select the atoms in an atomic beam,[17] and a natural method of velocity selection is

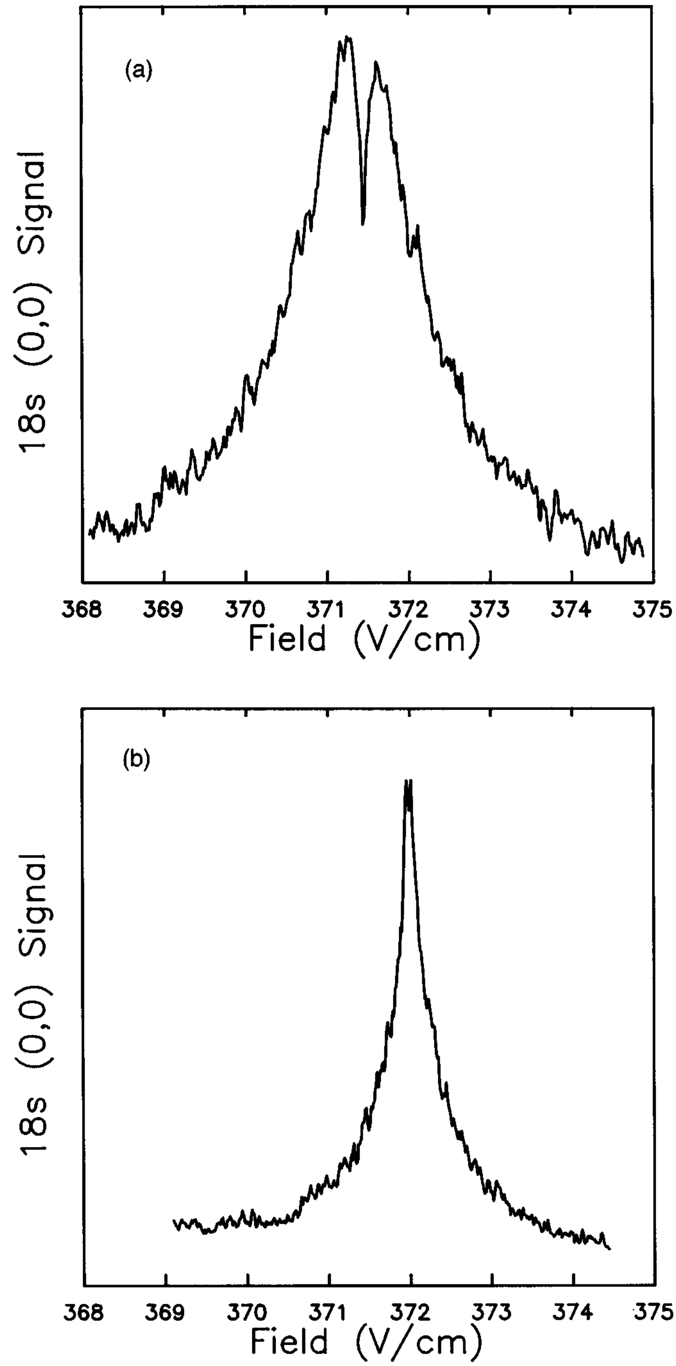

Fig. 14.12 Experimental measurements of the Na 18s (0,0) resonance for (a) $\mathbf{v} \parallel \mathbf{E}$ and (b) $\mathbf{v} \perp \mathbf{E}$. Note the definite dip near the center of the resonance in (a), as well as the slight asymmetry, both of which agree with the numerical predictions (from ref. 14).

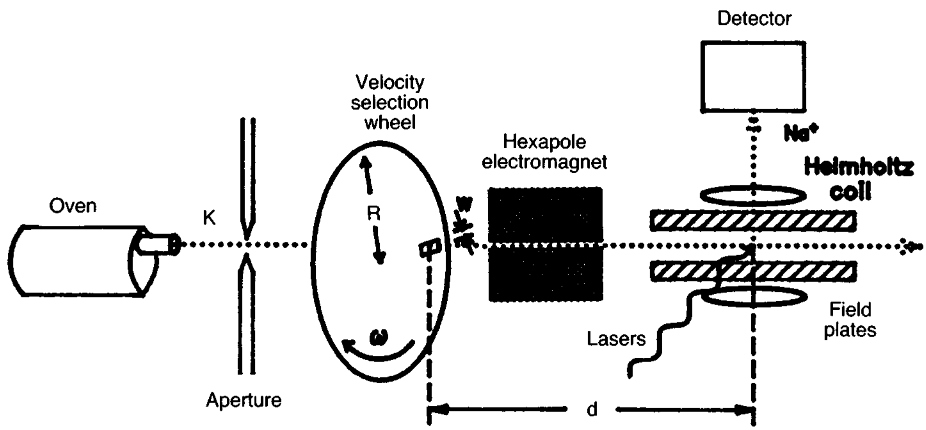

Fig. 14.13 Experimental arrangement for velocity selecting and focusing the atomic beam. The rotating slotted disc and the pulsed laser beam select atoms in a velocity group, and the hexapole magnet focuses them where they cross the laser beam (from ref. 18).

the one shown in Fig. 14.13.[18] Atoms effusing from the source pass through a chopper, a rotating 9.6 cm diameter disc which has radial slits as narrow as 1.5 mm near the outer edge. The disc rotates at frequencies as high as 200 Hz, at which frequency a 1.5 mm slit lets through a 25 μs wide pulse of atoms. The laser, which has a 5 ns pulse width, excites the atoms 250 μs later at the interaction region, which is 10 cm from the chopper. With a 1.5 mm wide slit the velocity distribution is ~10% of the velocity, which is usually chosen to be the velocity of the peak of the velocity distribution of the beam. Since a single velocity group of atoms is selected, it is possible to focus them spatially using the hexapole focusing magnet, substantially increasing the number density at the interaction region.[18,19]

The first and most obvious question is whether or not a narrower velocity distribution leads to narrower collisional resonances. In Fig. 14.14 we show the Na 26s + Na 26s $\rightarrow$ Na 26p + Na 25p resonances obtained under three different experimental conditions.[20] In Fig. 14.14(a) the atoms are in a thermal 670 K beam. In Figs. 14.14(b) and (c) the beam is velocity selected using the approach shown in Fig. 14.13 to collision velocities of 7.5×10^3 and 3.8×10^3 cm/s, respectively. The dramatic reduction in the linewidths of the collisional resonances is evident. The calculated linewidths are 400, 28, and 10 MHz, and the widths of the collisional resonances shown in Figs. 14.14(a)–(c) are 350, 40, and 23 MHz respectively. The widths decrease approximately as $1/v^{3/2}$ until Fig. 14.14(c), at which point the inhomogeneities of the electric field mask the intrinsic linewidth of the collisional resonance.

In the Na collision process of Eq. (14.1) the inhomogeneities in the static tuning fields required preclude the observation of very narrow collisional resonances. In

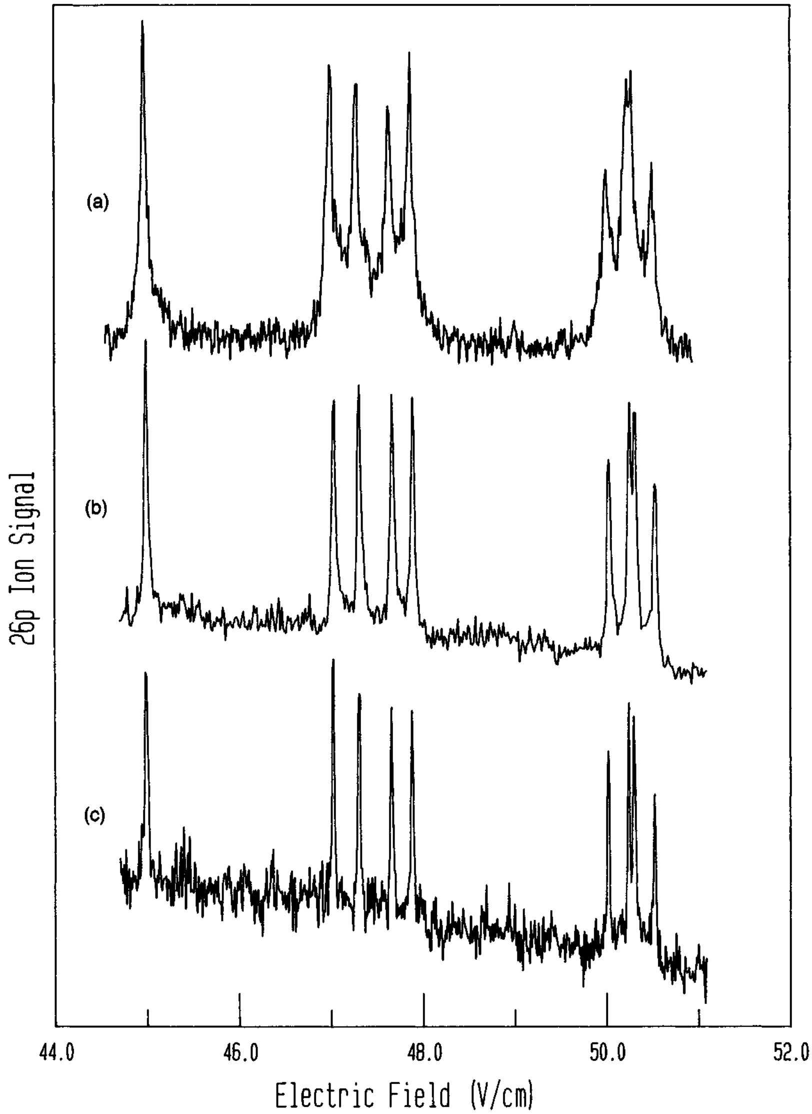

Fig. 14.14 Na 26s + Na 26s → Na 26p + Na 27p collisional resonances observed with (a) no velocity selection, collision velocity 4.6×10^4 cm/s, (b) velocity selection to a collision velocity 7.5×10^3 cm/s, (c) velocity selection to a collision velocity 3.8×10^3 cm/s (from ref. 20).

the K atom, however, dipole–dipole resonances occur near zero electric field,[18] where the homogeneity of the static field is less important. A K collision process which has been studied extensively is[17,18,21]

$$\text{K } 29s + \text{K } 27d \rightarrow \text{K } 29p + \text{K } 28p, \tag{14.30}$$

which occurs in relatively small fields as shown in the energy level diagram of Fig. 14.15. As shown by Fig. 14.15 and Eq. (14.30), the 29s and 27d states must both be

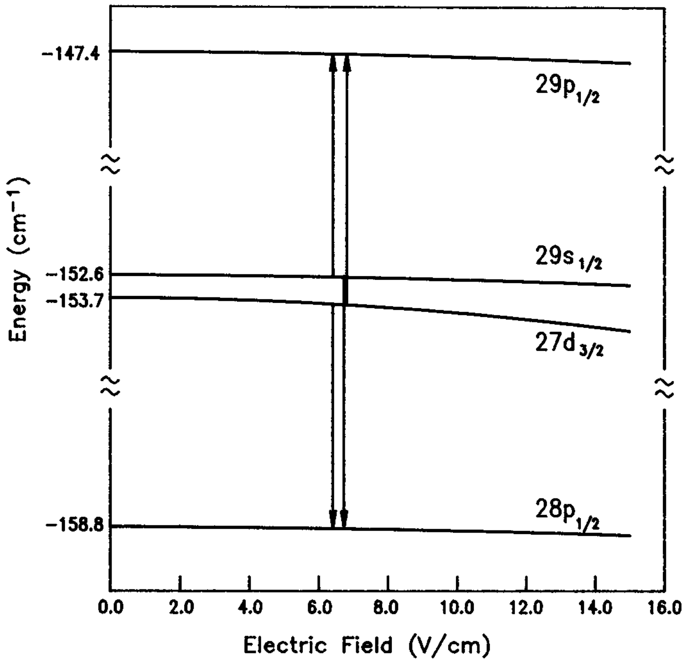

Fig. 14.15 Energy level diagram for the Rydberg states of K involved in the resonant energy transfer collisions being studied. The two transitions shown have been separated for clarity; they are degenerate in electric field (from ref. 21).

populated, and the fact that a resonant collision has occurred is ascertained by detecting the population in the higher lying 29p state by selective field ionization.

As shown by Fig. 14.15, the resonances occur near zero field, and it is easy to calculate the small Stark shifts with an accuracy greater than the linewidths of the collisional resonances. As a result it is straightforward to use the locations of the collisional resonances to determine the zero field energies of the *p* states relative to the energies of the *s* and *d* states. Since the energies of the *n*s and *n*d states have been measured by Doppler free, two photon spectroscopy,[22] these resonant collision measurements for $n = 27$, 28, and 29 allow the same precision to be transferred to the *n*p states. If we write the quantum defect δ_p of the K *n*p states as

$$\delta_p = \delta_p^0 + \delta_p^1(n^*)^{-2} + \delta_p^2(n^*)^{-4}, \tag{14.31}$$

these measurements have allowed a new determination of $\delta_p{}^0$ with a factor of 5 smaller uncertainty than the previous value.[23] Explicitly, they yield $\delta_p{}^0 = 1.711925(3)$, in which the major uncertainly is from the Doppler free laser spectroscopy, not from the resonant collision measurements.[17]

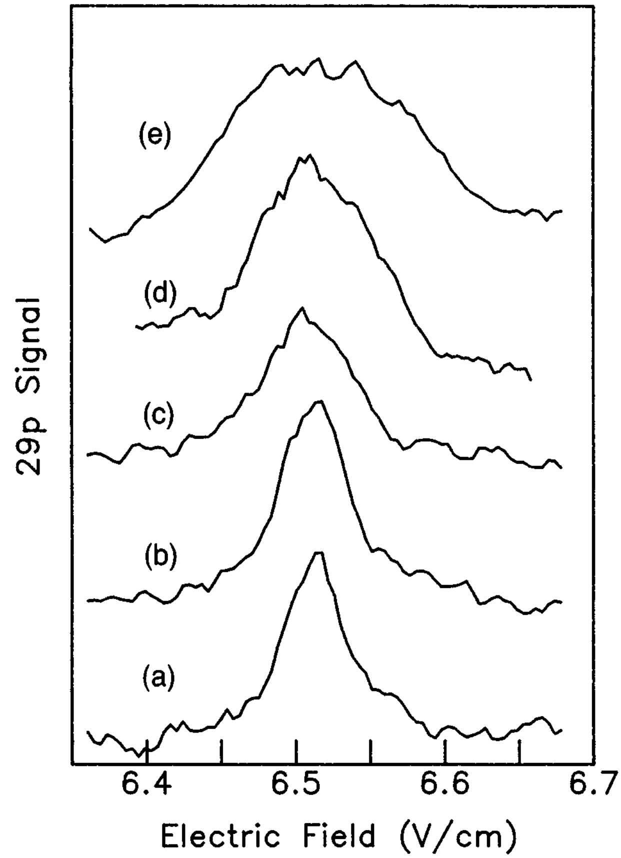

Fig. 14.16 Observed 29s resonance in K for several values of maximum-allowed collision time: (a) $\tau = 3.0\ \mu s$, FWHM = 1.4 MHz, (b), $\tau = 2.0\ \mu s$, FWHM = 1.8 MHz, (c) $\tau = 1.0\ \mu s$, FWHM = 2.2 MHz, (d), $\tau = 0.4\ \mu s$, FWHM = 3.1 MHz, and (e) $\tau = 0.2\ \mu s$, FWHM = 5.2 MHz (from ref. 21).

Transform limited collisions

While collisional resonances 5 MHz wide are interesting for spectroscopic purposes, what makes them most interesting is that the 5 MHz linewidth implies that the collision lasts at least 200 *n*s, a time not much less than the 1 μs period allowed for the collisions to occur. If the collision linewidths can be reduced to the inverse of the time allowed for the collisions to occur, the collisional resonances become transform limited, and we know when each collision begins and ends.

Thomson *et al.* reached the transform limit by incorporating two improvements.[21] First, Helmholtz coils were placed around the interaction region to cancel the earth's magnetic field. Second, the time interval allowed for the collisions had previously been defined by the laser pulse and the field ionization pulse, which rises slowly at its onset. In these experiments the interval was terminated by the application of a rapidly rising, ~50 ns, low voltage detuning pulse which switches the field away from the field value of the collisional resonance at a well defined time.[24] These improvements allowed the observation of collisional resonances as narrow as 1 MHz. These narrow resonances were then transform broadened by reducing the time the atoms are allowed to collide to below 1 μs. In Fig. 14.16 we show collisional resonances observed as the time between the laser pulse and the detuning pulse is shortened from 1 μs to 0.4 μs, showing clearly the transform broadening of the collisional resonances as the time interval is shortened. If the collisional resonances are transform limited, we know when individual collisions begin and end, and we can perturb the colliding atoms at well defined times during the collision, for example, when the atoms are at their point of closest approach.

References

1. C. K. N. Patel, in *Lasers*, Vol. 2, ed. A. K. Levine (Marcel Dekker, New York, 1968).
2. A. Javan, W. R. Bennet, Jr., and D. R. Herriot, *Phys. Rev. Lett.* **8**, 470 (1962).
3. A. D. White and J. D. Rigden, *Proc. I.R.E.* **50**, 9167 (1962).
4. C. K. N. Patel, *Phys. Rev. Lett.* **13**, 617 (1964).
5. T. F. Gallagher, *Phys. Rept.* **210**, 319 (1992).
6. P. L. Houston, in *Advances in Chemical Physics*, Vol. 47, eds. I. Prigogine and S.A. Rice (Wiley, New York, 1981).
7. T. Oka, in *Advances in Atomic and Molecular Physics*, Vol. 9, eds. D. R. Bates and I. Esterman (Academic, New York, 1973).
8. P. W. Anderson, *Phys. Rev.* **76**, 647 (1949).
9. K. A. Smith, F. G. Kellert, R. D. Rundel, F. B. Dunning, and R. F. Stebbings, *Phys. Rev. Lett.* **40**, 1362 (1978).
10. T. F. Gallagher, G. A. Ruff, and K. A. Safinya, *Phys. Rev. A* **22**, 843 (1980).
11. K. A. Safinya, J. F. Delpech, F. Gounand, W. Sandner, and T. F. Gallagher, *Phys. Rev. Lett.* **47**, 405 (1981).
12. T. F. Gallagher, K. A. Safinya, F. Gounand, J. F. Delpech, W. Sandner, and R. Kachru, *Phys. Rev. A* **25**, 1905 (1982).
13. E. Fiordilino, G. Ferrante, and B. M. Smirnov, *Phys. Rev. A* **35**, 3674 (1987).
14. D. S. Thomson, R. C. Stoneman, and T. F. Gallagher, *Phys. Rev. A* **39**, 2914 (1989).
15. N. F. Ramsey, *Molecular Beams* (Oxford University Press, London, 1956).
16. D. S. Thomson, PhD Thesis, University of Virginia (1990).
17. R. C. Stoneman, M. D. Adams, and T. F. Gallagher, *Phys. Rev. Lett.* **58**, 1324 (1987).
18. M. J. Renn and T. F. Gallagher, *Phys. Rev. Lett.* **67**, 2287 (1991).
19. A. Lemonick, F. M. Pipkin, and D. R. Hamilton, *Rev. Sci. Instr.* **26**, 1112 (1955).
20. M. J. Renn, R. Anderson, and Q. Sun, private communication (1992).
21. D. S. Thomson, M. J. Renn, and T. F. Gallagher, *Phys. Rev. Lett.* **65**, 3273 (1990).
22. D. C. Thompson, M. S. O'Sullivan, B. P. Stoicheff, and Gen-Xing Xu, *Can. J. Phys.* **61**, 949 (1983).
23. P. Risberg, *Ark. Fys.* **10**, 583 (1956).
24. T. H. Jeys, K. A. Smith, F. B. Dunning, and R. F. Stebbings, *Phys. Rev. A* **23**, 3065 (1981).